

Infrared Thermal-Guided Adversarial Patch Defense for Robust Visible Person Detectors

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Abstract

Adversarial patch attacks on visible person detectors pose significant risks to recognition and safety. To counter this, we propose a defense method that integrates thermal imaging with RGB images to exploit differences in thermal characteristics between adversarial patches and humans. Our approach segments thermal images using selective search to generate mask sets and apply them to RGB images for attack detection and mitigation. To evaluate the proposed method, we created a physical attack dataset by conducting a hiding attack using printed adversarial patches against a person detector. This dataset enabled a quantitative assessment of our method. Experimental results demonstrate that our approach effectively defends against adversarial patches while maintaining performance on clean images, validating its robustness in practical scenarios.

1. Introduction

Deep learning-based object detection is a branch of computer vision that classifies and locates objects in images and videos. Unlike image classification, which only identifies objects, object detection utilizes bounding boxes for precise localization. It is widely applied in the real-world, particularly for person detection, which is essential for safety, security, and efficiency [1]. For example, autonomous vehicles detect pedestrians for navigation, while security CCTV systems monitor and detect unauthorized entries. As object detectors become more prevalent, analyzing their vulnerabilities is critical. Visible object detectors are particularly susceptible to adversarial patches [3, 16], which manipulate image regions to deceive models, potentially leading to severe consequences. In surveillance, such patches can

prevent the detection of critical objects like persons or vehicles [21, 13], while in autonomous driving, they may cause the system to overlook pedestrians, leading to real-world harm [23]. These attacks are especially hazardous because they can be physically deployed [19, 23]. Previous studies have shown that adversarial patches can mislead object detectors in the world [2, 12], and even simply holding such patches can hide a person from detection [18, 21]. Therefore, developing robust defenses that can detect and mitigate adversarial patches is crucial.

We focus on defending against adversarial patches that target visible object detectors using RGB cameras. Since RGB cameras struggle in low-light conditions, they are often complemented by thermal cameras in systems like CCTV and autonomous vehicles. Thermal cameras capture heat signatures, allowing them to provide additional information about objects. Since adversarial patches typically do not emit heat, they are expected to exhibit distinct thermal properties compared to humans, suggesting that thermal imaging could help distinguish them. Motivated by this, we explore methods to detect and mitigate adversarial patches using thermal information.

Our proposed method leverages selective search [22] to segment thermal images, allowing us to identify and mitigate adversarial patches. First, thermal images are segmented based on thermal intensity, object size, and temperature distribution, then gradually merged to highlight object regions. From these segmented regions, we generate mask sets and apply them to the RGB image. By analyzing changes in detection results before and after masking, we detect adversarial patches. If a person remains undetected initially but appears after masking, this suggests the patch is interfering with detection. Also, to mitigate this effect, we select the masked image with the highest confidence score and use it for final detection. To validate our defense method, we conducted a hiding attack using an adversarial patch on a person detector, preventing it from recognizing the object by reducing the model's objectness score. We first generated adversarial patches in the digital environment and printed them

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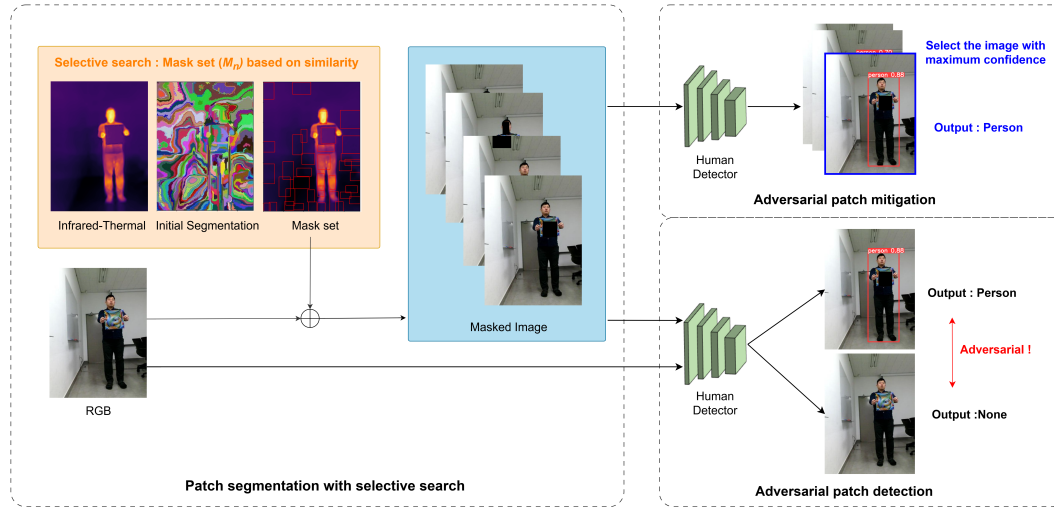


Figure 1. Our proposed defense method uses infrared-thermal segmentation to mask the RGB image. (Top) Adversarial patch mitigation by selecting the image with the maximum confidence among the original and masked images; (Bottom) Adversarial patch detection by comparing the person detector’s output label between the original and masked images.

on paper for physical attacks. Due to the limited availability of physical attack data, we created a dedicated dataset by photographing our lab members holding the printed patches. The proposed defense method was then evaluated using this dataset, with detailed information provided in Section 4. Since few studies have presented quantitative experimental results for physical attacks, our work contributes by providing comprehensive evaluations of both physical attacks and defenses, demonstrating the robustness of our approach. In summary, the main contributions of this work are as follows.

- We propose an infrared thermal-guided defense that enhances the robustness of person detectors against adversarial patches through mitigation and detection.
- We empirically demonstrate effective defense against adversarial patch attacks while maintaining performance on clean images, ensuring reliability.
- The proposed approach is model-agnostic, requiring no modifications to the detector’s architecture, making it adaptable to various person detection systems.

2. Background and related work

2.1. Adversarial patch attack

An adversarial patch is a localized attack that manipulates a prediction of the model by altering specific image regions. By physically placing patches on objects, attackers can deceive target models, posing significant threats to real-world applications such as autonomous driving and surveillance. Initially designed for image classification, adversarial patches have since evolved to threaten object detection and other vision-based tasks. Brown *et al.* [3] introduced a universal patch that misled classification models regardless of

its position or size, while Eykholt *et al.* [8] demonstrated real-world attack by applying black-and-white stickers to traffic signs, causing misclassification. Liu *et al.* [16] proposed Dpatch, the first adversarial patch against object detection, enabling targeted and untargeted attacks by degrading bounding box predictions. Eykholt *et al.* [20] extended these attacks to the physical world, making objects vanish or generating false detections. Thys *et al.* [21] targeted surveillance models with an adversarial patch attack on person detection, addressing significant intra-class variations. Subsequent studies have further refined adversarial patch techniques for increased robustness in practical applications.

2.2. Defenses against adversarial patch attacks

Defense strategies against adversarial patch attacks include mitigation, detection, masking, and restoration through inpainting. Dziugaite *et al.* [7] proposed JPEG compression (JPEG) to remove high-frequency noise, reducing subtle pixel changes from adversarial patches and enhancing robustness. Its simplicity makes it a popular choice for enhancing computer vision. Chen *et al.* [4] introduced Jujutsu, which uses saliency maps to mask adversarial patches for detection and mitigation. However, these methods mainly target classification tasks. Xiang and Mittal [24] were the first to introduce DetectorGuard, a defense mechanism for object detectors against hiding attacks. This method uses an objectness predictor to estimate object presence and compare it with the base detector’s results to detect adversarial patch attacks and issue alerts. Ji *et al.* [9] developed Ad-Yolo, which can detect adversarial patches as if they were another object by training adversarial patches. However, although it effectively detects the presence of patches, it does not mitigate the

attack's impact by restoring accurate person detection. Liu *et al.* [15] proposed Segment and Complete (SAC), which includes a patch segmenter that integrates a robust shape completion algorithm with adversarial training to effectively remove patches. However, these methods require generating adversarial images beforehand and additional training.

3. Proposed method

Our method leverages thermal imaging to mitigate and detect adversarial patch attacks. Modern person detection systems integrate infrared thermal and RGB imaging for robustness under varying lighting conditions. Based on this, we use thermal imaging to identify adversarial patches in RGB images. Adversarial patches are generated digitally and then printed on paper, as they effectively retain color and size while being cost-effective and accessible. Unlike human skin, they do not emit heat, allowing thermal cameras to capture temperature differences for detection.

The overall defense framework is illustrated in Figure 1. First, thermal images corresponding to RGB images are segmented using selective search and merged based on their characteristics to generate mask sets likely to contain adversarial patches. These masks are used for attack mitigation and detection. For attack mitigation, we select the masked image with the highest confidence score and use it as input, reducing the patch's impact and restoring object detection performance. For attack detection, we compare the object detection results of the original image with those of the masked image. If the results differ, the image is detected as an adversarial patch attack.

3.1. Problem formulation

Object detector We use YOLOv5 [10] as the target model, a widely used one-stage detector known for its efficiency and accuracy. The model takes an image as input and outputs bounding box coordinates, an objectness score representing the likelihood of an object's presence, a confidence score for class probability, and the predicted class. To remove duplicate detections, Non-Maximum Suppression (NMS) retains the highest-confidence bounding box and suppresses overlaps above an IoU threshold.

Attack formulation We conduct a hiding attack, which prevents object detectors from recognizing the presence of persons and disrupts their primary function. Thus, we create adversarial patches that reduce the objectness score below a threshold while considering various physical factors. These patches are designed using the objective function:

$$P(x, l) = \arg \min_P \mathbb{E}_{x, t, l} [L(f(A(x, P, l, t)))] \quad (1)$$

where f is the object detector, x is the image, P is the patch, l is the location, t denotes the transformations of the patch, and $A(x, P, l, t)$ denotes applying the patch to the image.

Algorithm 1 Patch Segmentation

Input: Thermal Image T , Initial Region K , Top N Regions, IoU Threshold τ , Size Range $[s_{\min}, s_{\max}]$

Output: Mask Set M

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1:  $S \leftarrow \text{SelectiveSearch}(T, K, N)$ 
2:  $S_{\text{filtered}} \leftarrow \{s \in S \mid s_{\min} \leq \text{size}(s) \leq s_{\max}\}$ 
3:  $M \leftarrow \{s_i \in S_{\text{filtered}} \mid \text{IoU}(s_i, s_j) \leq \tau, \forall s_j \neq s_i\}$ 
4: function SelectiveSearch( $T, K, N$ )
5:    $T_{\text{Lab}} \leftarrow \text{Convert thermal image to Lab color space}$ 
6:    $S \leftarrow \text{Initial segmentation with region } K \text{ from } T_{\text{Lab}}$ 
7:   while  $S \neq \emptyset$  do
8:      $S_i, S_j \leftarrow \arg \max_{(i, j) \in S} \text{Sim}(S_i, S_j)$ 
9:      $S \leftarrow (S \cup \text{Merge}(S_i, S_j)) \setminus S_i, S_j$ 
10:  end while
11:   $R \leftarrow \text{Select top } N \text{ regions by similarity from } S$ 
12:   $S \leftarrow \text{Generate mask for } r \in R$ 
13: end function

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The loss function L is a weighted sum of four components: Non-Printability Score (L_{NPS}) [19], Total Variation loss (L_{TV}) [17], Saliency loss (L_{sal}) [6], and Objectness loss (L_{obj}) [21]. L_{NPS} ensures the patch's colors are printable by minimizing differences from standard printer-reproducible RGB triplets; L_{TV} smooths the patch by reducing high-frequency pixel variations; L_{sal} lowers color contrast by minimizing the standard deviation between color channels, making the patch less noticeable; and L_{obj} is minimized to reduce the objectness score for person detection:

$$L = \alpha L_{\text{NPS}} + \beta L_{\text{TV}} + \gamma L_{\text{sal}} + \delta L_{\text{obj}}, \quad (2)$$

where α , β , γ , and δ are hyperparameters.

3.2. Patch segmentation with selective search

We segment infrared thermal images into similar regions to detect adversarial patches based on their thermal contrast. While various materials can be used for adversarial patches, we focus on printed paper due to its practicality and effectiveness in prior studies [12, 2, 20]. Differences in thermal properties, such as conductivity and emissivity, make materials distinguishable and provide key cues for segmentation, though effectiveness may vary depending on the material.

We use selective search [22] for patch segmentation, known for its effectiveness in detecting objects with distinct features in complex backgrounds. Algorithm 1 provides the procedure for applying selective search to patch segmentation. Segmentation relies on three similarity measures: color, size, and fill. Color represents thermal intensity, size indicates the extent of the object, and fill reflects density and uniformity. The similarity $\text{Sim}(i, j)$ between regions i and j is computed from these elements. The resulting rectangular segments form a mask set M , serving as candidate regions for adversarial patches. To refine segmentation, we filter

the initial mask set S based on size constraints and remove overlapping regions using an IoU threshold τ , improving precision and processing speed by reducing candidates.

3.3. Infrared thermal-guided based adversarial patch defense

Adversarial patch mitigation The mitigation process minimizes the impact of adversarial patches on input images when their presence is uncertain. Similar to detection, the image is masked using the mask set of patch segmentation. The system selects the image with the highest confidence score from either the original or masked images as the final input. This process mitigates potential attack effects so the person detector functions correctly, enhancing attack resilience. While this approach mitigates attack effects, it is also essential to preserve the person detection performance on clean images, ensuring reliable operation.

Adversarial patch detection The detection process involves comparing the person detector results before and after applying the mask set. If an adversarial patch is present, the person detector fails to detect the person. However, masking suspected patch regions removes its influence, enabling the person to be correctly detected in the masked image. If a person is undetected in the original image but detected after masking, it indicates an adversarial attack. This process alerts the system to potential threats, enabling countermeasures against intentional evasion.

4. Experiments

4.1. Adversarial patch attack

Dataset We used the INRIA Person (INRIA)[5] and the Microsoft Common Objects in Context (COCO)[14] datasets. INRIA contains 614 training and 288 test images primarily for pedestrian detection, while COCO contains 118,000 training and 5,000 validation images focused on the “person” category. To evaluate physical-world attacks, we printed the optimized patches on paper and collected a physical dataset from 20 participants performing natural movements (e.g., raising and lowering arms). Images were captured using a FLIR ONE Edge Pro, recording paired RGB and infrared thermal images. Due to differences in angles and resolutions, the images were aligned and resized to 640×480. Annotations were generated using Roboflow’s auto-labeling with Grounding DINO and thoroughly manually verified by the authors, ensuring high annotation accuracy and quality.

Implementation details The target model includes YOLOv5_{INRIA}, trained on INRIA (512×512, 100 epochs), and YOLOv5_{COCO}, pre-trained on COCO. Adversarial patches were optimized using all datasets with $\alpha = 0.1$, $\beta = 2.5$, $\gamma = 1.0$, and $\delta = 3.0$, and enhanced for physical robustness through random transformations (e.g. contrast, brightness, rotation). The patch was positioned within $\pm 25\%$



Figure 2. Result of adversarial patch attacks in the physical world.

of the person’s center at size ratios of 0.1 to 0.3, with the most successful one over 1000 epochs being selected.

Digital to physical patch In digital attacks, person detectors reliably identified individuals without adversarial patches, but detection confidence and accuracy of the bounding box dropped significantly when applied. In physical attacks, participants held and moved with printed patches while the detector processed RGB images from the FLIR camera. An attack was deemed successful if the confidence fell to 0.7 or below, indicating failure to detect the person.

Physical attack results We evaluate attacks using Precision (P), Recall (R), and Average Precision (AP). Precision is the proportion of correct detections among all detections, while recall is the proportion of correctly detected objects among actual objects. AP is computed as the mean across IoU thresholds from 0.50 to 0.95 in increments of 0.05. In clean, precision and recall are 1.000 for both models. However, under attack, they drop to 0.962/0.885 for YOLOv5_{INRIA} and 0.954/0.919 for YOLOv5_{COCO}. Similarly, AP decreases significantly, with YOLOv5_{INRIA} dropping from 0.788 to 0.601 and YOLOv5_{COCO} from 0.957 to 0.781. Figure 2 visually illustrates the attack impact. The adversarial patch prevents person detection by reducing confidence scores and eliminating the bounding box.

4.2. Defense evaluation of physical patch attacks

Dataset For defense evaluation, we created a dataset containing 1,648 images: 271 clean images without adversarial patches, and 1,377 attack images from successful patch attacks on YOLOv5_{INRIA} and YOLOv5_{COCO} (814 and 563 images, respectively). These attack images were collected during the physical attacks described in Section 4.1, where the attacks successfully fooled the person detectors.

Baseline and implementation details We compared our method against two existing defenses: JPEG [7] and SAC [15]. JPEG is easy to implement and effectively reduces noise in images without requiring prior knowledge. SAC detects and removes patches using a patch segmenter trained through adversarial training. We used two versions of SAC: SAC_c, trained on COCO digital patches, and SAC_a, trained on the Apricot physical patch dataset [2]. Our method employs selective search segmentation on infrared thermal images to generate masks from extracted bounding boxes.

Target Model	Method	Clean					Attack					Mean
		P	R	AP ₅₀	AP ₇₅	AP	P	R	AP ₅₀	AP ₇₅	AP	
YOLOv5 _{INRIA}	Undefend	1.000	1.000	0.995	0.931	0.789	0.944	0.711	0.920	0.567	0.564	0.658
	JPEG [7]	1.000	1.000	0.995	0.933	0.815	0.952	0.705	0.931	0.578	0.572	0.672
	SAC _a [15]	0.977	0.952	0.974	0.810	0.675	0.999	0.997	0.990	0.987	0.847	0.752
	SAC _c [15]	1.000	1.000	0.995	0.939	0.756	0.944	0.721	0.925	0.569	0.566	0.648
	Ours	1.000	1.000	0.995	0.943	0.803	0.977	0.982	0.993	0.775	0.677	<u>0.735</u>
YOLOv5 _{COCO}	Undefend	1.000	1.000	0.995	0.995	0.965	0.902	0.892	0.956	0.800	0.684	<u>0.802</u>
	JPEG [7]	1.000	1.000	0.995	0.995	0.967	0.890	0.840	0.930	0.763	0.656	0.782
	SAC _a [15]	0.917	0.943	0.965	0.671	0.632	0.989	0.998	0.995	0.992	0.870	0.733
	SAC _c [15]	1.000	1.000	0.995	0.995	0.963	0.898	0.912	0.959	0.822	0.684	0.801
	Ours	1.000	1.000	0.995	0.995	0.966	0.945	0.957	0.955	0.833	0.710	0.819

Table 1. Comparison of attack mitigation results against clean and adversarial patches across methods. **Best** are bold; second underlined.

Target Model	Clean Accuracy	Attack Detection Rate
YOLOv5 _{INRIA}	100%	84.2%
YOLOv5 _{COCO}	99.3%	83.1%

Table 2. Attack detection results using the proposed method.

The thermal images are converted to Lab color space and segmented based on color, size, and fill similarity. Segmentation is initialized with $K = 200$, selecting the top 100 regions. Bounding boxes are filtered within a size range of 30 to 100, and an IoU threshold of 0.20 is applied to remove overlaps.

Metrics We evaluated our attack mitigation method using precision (P), recall (R), AP₅₀, AP₇₅, AP, and the harmonic mean of AP for both clean and attack scenarios. The attack detection performance was measured by clean accuracy (the proportion of correctly identified clean) and attack detection rate (the proportion of correctly identified adversarial).

Mitigation performance against adversarial patches

Table 1 compares defense methods against clean and adversarial patch attacks. JPEG maintains performance on clean images, with some metrics even improving slightly due to noise reduction. However, its defense against adversarial patches is minimal, showing only minor variations in performance without a consistent improvement. SAC_a significantly reduces clean image performance, dropping AP from 0.789 to 0.675 in YOLOv5_{INRIA} and from 0.965 to 0.632 in YOLOv5_{COCO}. However, it effectively mitigates adversarial patches, increasing AP from 0.564 to 0.847 in YOLOv5_{INRIA} and from 0.684 to 0.870 in YOLOv5_{COCO}. Consequently, this suggests that while SAC_a provides effective defense capabilities, it significantly reduces the model’s performance on clean images. SAC_c maintains performance on clean images but offers little improvement against adversarial attacks, with only a 0.002 increase in AP.

Our proposed method slightly enhances clean image performance and demonstrates robust defense against adversarial patches, improving AP to 0.677 (+0.143) in YOLOv5_{INRIA} and 0.710 (+0.026) in YOLOv5_{COCO}. Evaluating overall performance using mean AP, SAC_a ranks second

with 0.752 for YOLOv5_{INRIA}, while our method achieves the highest score of 0.819 for YOLOv5_{COCO}. Although a trade-off typically exists between clean and adversarial image performance, our method outperforms others in balancing both aspects. Figure 3 illustrates how our method detects adversarial attacks and effectively restores person detection.

Detection performance against adversarial patches

Table 2 presents the detection rates using our method. For YOLOv5_{INRIA}, clean images are correctly identified as non-attacks 100%, with no false positives, while adversarial patches are detected at a rate of 84.2%. For YOLOv5_{COCO}, clean images are correctly classified 99.3%, with a 0.7% false positive rate, and adversarial patches are detected at 83.1%. The high clean accuracy minimizes false alarms, while the strong detection rates highlight the effectiveness of our method in identifying adversarial patches.

Defense against transferable patch on YOLOv8

We extended our evaluation to YOLOv8n, the lightweight version of YOLOv8 [11], to assess the applicability of our defense method. Using the physical patch attack dataset from YOLOv5, we conducted transfer attacks on YOLOv8n.

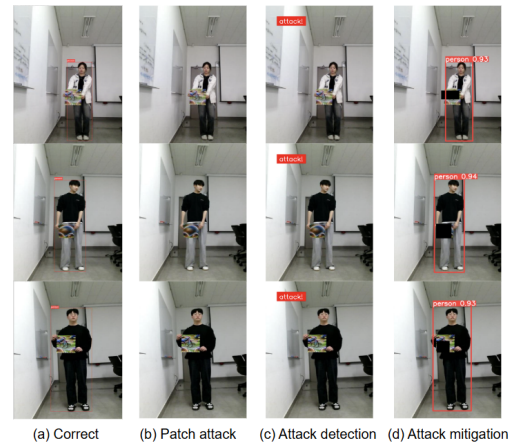


Figure 3. Detection examples using the proposed method: (a) Correct detection, (b) Person hidden by adversarial patch, (c) Attack detection, (d) Person detection restored after attack mitigation.

Target Model	Dataset	P	R	AP ₅₀	AP ₇₅	AP
YOLOv8n	Clean	1.000	0.995	0.995	0.995	0.988
	Attack	0.876	0.791	0.910	0.664	0.637
	Ours	0.953	0.871	0.955	0.804	0.791

Table 3. Attack mitigation results on YOLOv8n.

The results showed that some adversarial effects persisted, leading to a transfer attack success rate of 8.4%, demonstrating that adversarial patches can still effectively compromise model performance even when not optimized for the target architecture. As shown in Table 3, the attack significantly degraded YOLOv8n’s performance compared to the clean. Our defense method effectively mitigated the attack impact of the adversarial patch. This suggests that the proposed defense is generalizable to newer object detection architectures.

5. Conclusion and future work

We propose an infrared thermal-guided method for mitigating and detecting adversarial patch attacks in the physical world. Our approach segments thermal images to generate mask sets, which are then applied to RGB images. The attack mitigation selects the most reliable masked image to counteract the effects of the attack, and detection identifies adversarial patches. Our evaluation demonstrates that the method preserves person detection performance on clean images while mitigating the impact of adversarial patches, proving robustness compared to existing defenses. However, we tested our approach using paper-based adversarial patches, and the effectiveness of the method may vary depending on the thermal properties of patch materials. Future work should explore generalizing the model to diverse materials, outdoor environments, and illumination conditions, with particular emphasis on optimizing the segmentation step for real-time applications.

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